Thermodynamics – heat, work and entropy

Thermodynamics is the only part of physics created by engineers: it grew out of the question of how much work can be got out of a steam engine. The answer became four laws that apply to everything from coffee cups to black holes, without any need to know about atoms. Einstein wrote that it was the only physical theory of universal content which he was convinced would never be overthrown. This page covers the laws, the cycles and phase transitions with two interactive figures.

The zeroth law and temperature

If two bodies are each in thermal equilibrium with a third, they are in equilibrium with each other. The law was first stated explicitly by Ralph Fowler around 1930 – after the other three – but logically it is the precondition for a thermometer to make sense: temperature is the quantity that is the same for bodies in equilibrium.

Daniel Fahrenheit (1724), Anders Celsius (1742) and others defined practical scales from fixed points. The ideal-gas thermometer showed that all gases at low pressure point to the same absolute zero at −273.15 °C. In 1848 William Thomson (Lord Kelvin) defined an absolute scale that depends not on any particular substance but on Carnot's cycle. Since 2019 the kelvin has been defined by fixing Boltzmann's constant: $k_B = 1.380\,649 \cdot 10^{-23}$ J/K exactly.

The first law: energy is conserved

Around 1800 most people thought of heat as an invisible substance, *caloric*. In 1798 in Munich Benjamin Thompson (Count Rumford) noticed that boring cannon could produce unlimited heat as long as the horses kept pulling. Julius Robert Mayer (1842), James Prescott Joule (1843–50) and Hermann von Helmholtz (1847) established that heat and work are two forms of energy. Joule measured the mechanical equivalent of heat by letting a falling weight drive a paddle wheel in water: 4.18 J per calorie. As a law:

$$\Delta U = Q - W$$

The change in a system's internal energy is the heat supplied minus the work done by the system. For an expanding gas the work is $W = \int p\,dV$ – the area under the curve in a pV diagram. U is a function of state that depends only on the state of the system; Q and W depend on the path.

The ideal gas and its processes

Boyle's law (1662), Charles's law (1787) and Gay-Lussac's law (1802) were combined by Émile Clapeyron in 1834 into the ideal gas equation $pV = nRT$. For an ideal gas the internal energy depends only on temperature, $U = nC_V T$, and Mayer found the relation $C_p - C_V = R$ for molar heat capacities. The four standard processes:

Process	Constant	Work W	Heat Q
Isochoric	V	0	$nC_V\Delta T$
Isobaric	p	$p\,\Delta V$	$nC_p\Delta T$
Isothermal	T	$nRT\ln(V_2/V_1)$	$= W$
Adiabatic	pV^γ, $\gamma = C_p/C_V$	$-nC_V\Delta T$	0

The second law: heat flows from hot to cold

In 1824 the 28-year-old French engineer Sadi Carnot published *Réflexions sur la puissance motrice du feu*. He did not ask how a particular engine could be improved, but what the greatest efficiency of any heat engine could be. The answer: it depends only on the temperatures T_h and T_c of the hot and cold reservoirs, not on the working substance. Carnot still believed in caloric, but the argument held. He died of cholera in 1832, and the book only became known when Clapeyron and Kelvin took it up.

$$\eta_{\max} = 1 - \frac{T_c}{T_h}$$

Rudolf Clausius (1850) and Kelvin (1851) stated the law in two equivalent ways:

Clausius

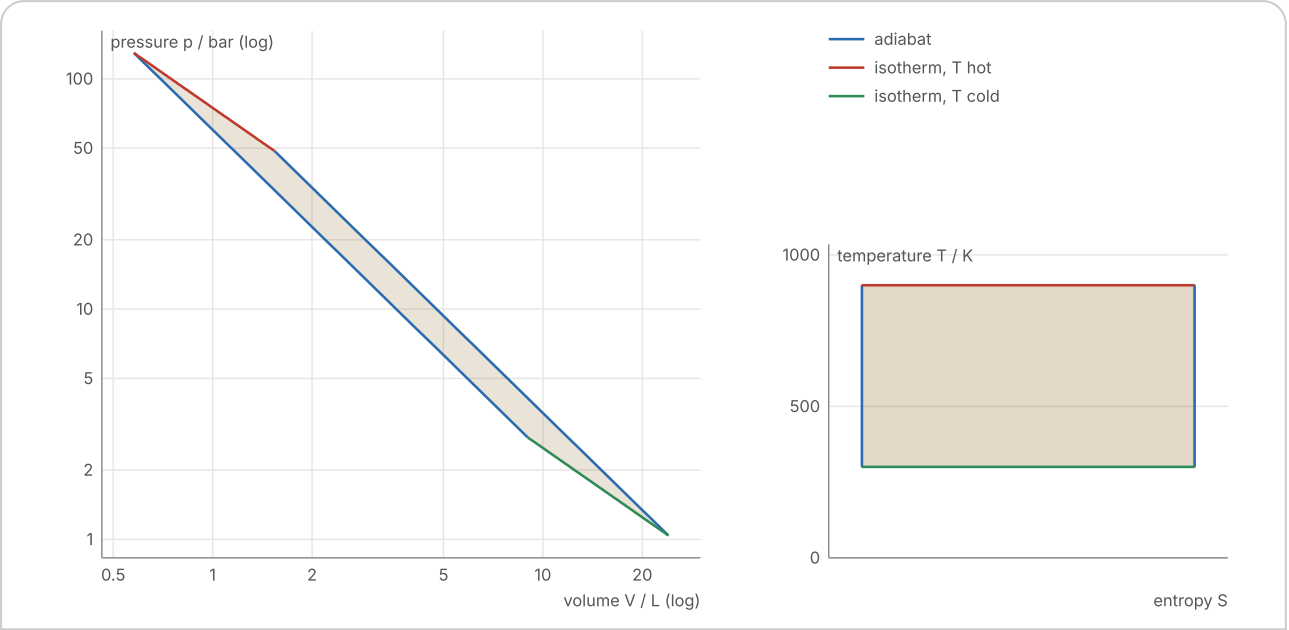
Heat cannot of itself flow from a colder to a hotter body.

Kelvin–Planck

No cyclic process can convert heat from a single reservoir completely into work.

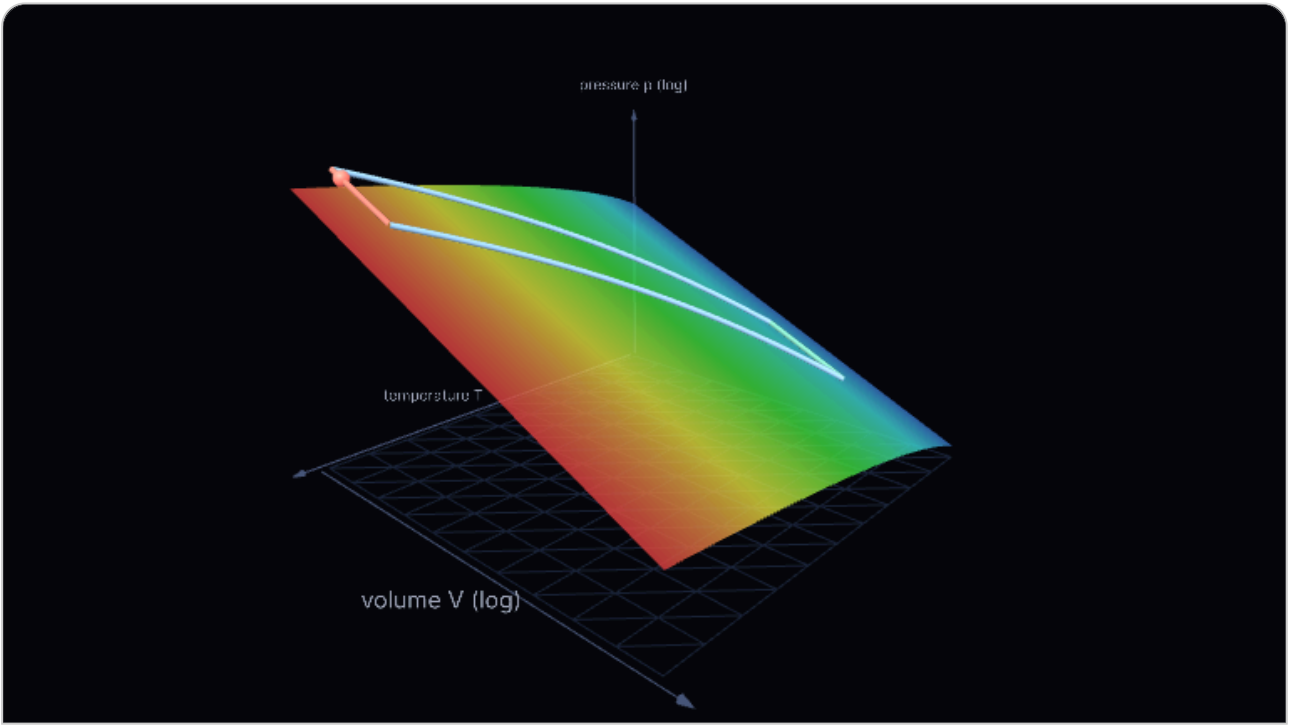
A power station working between 550 °C and 30 °C can have an efficiency of at most $1 - 303/823 \approx 63\,\%$; in practice the best reach about 45–47 %.

Cycles



See also the animated pages on the Carnot cycle, the Stirling engine, the steam locomotive and turbines.

The ideal gas surface (3D)



Entropy

In 1854–65 Clausius realised that Carnot's result means that the quantity $\oint dQ/T$ is zero for any reversible cycle. So there must exist a function of state whose change is $dS = dQ_{\text{rev}}/T$. He

called it *entropy* after the Greek word for transformation and summed up thermodynamics in two sentences: The energy of the world is constant. The entropy of the world tends towards a maximum.

$$dS \geq \frac{dQ}{T}, \qquad \Delta S_{\text{isolated}} \geq 0$$

In 1877 Ludwig Boltzmann gave entropy a microscopic interpretation: $S = k_B \ln W$, where W is the number of microscopic states corresponding to the macroscopic state. Entropy increases because disordered states are overwhelmingly more probable. See Statistical mechanics, Entropy and The arrow of time.

Thermodynamic potentials and Maxwell relations

Combining the first and second laws for reversible processes gives the fundamental relation $dU = T\,dS - p\,dV$. By Legendre transformations (Josiah Willard Gibbs, 1873–78) one can switch to other natural variables:

Potential	Definition	Differential	Minimum at constant
Internal energy	U	$T\,dS - p\,dV$	S, V
Enthalpy	$H = U + pV$	$T\,dS + V\,dp$	S, p
Helmholtz free energy	$F = U - TS$	$-S\,dT - p\,dV$	T, V
Gibbs free energy	$G = H - TS$	$-S\,dT + V\,dp$	T, p

Because the mixed second derivatives are equal, one obtains the Maxwell relations, e.g. $\left(\partial S/\partial V\right)_T = \left(\partial p/\partial T\right)_V$. They connect quantities that are hard to measure (entropy) with ones that are easy (pressure and temperature). The Gibbs free energy governs chemical equilibria – see Le Chatelier's principle and Gibbs' paradox.

The third law

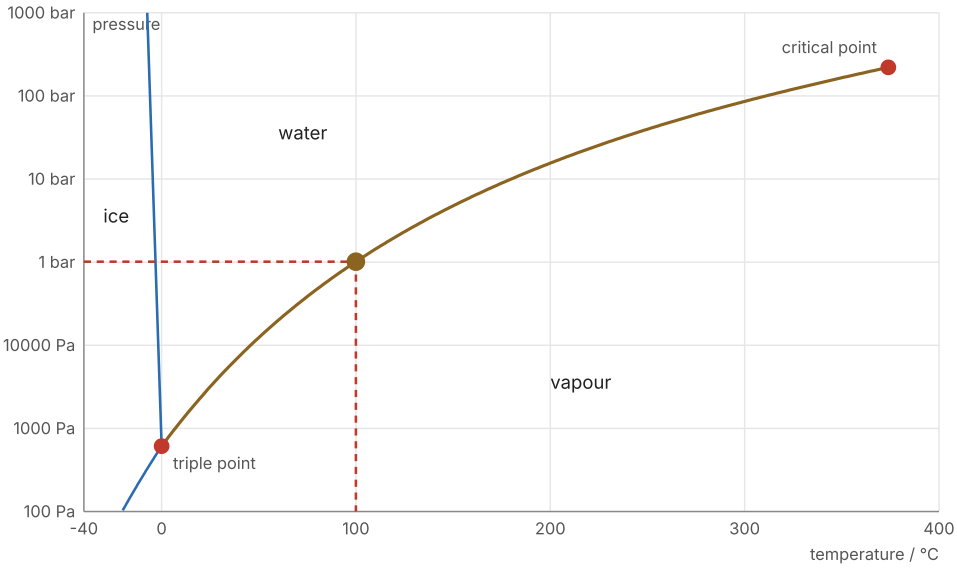
In 1906 Walther Nernst found, from measurements of chemical reactions at low temperatures, that entropy changes go to zero as the temperature approaches absolute zero. Max Planck sharpened this: the entropy of a perfect crystalline substance is zero at 0 K. One consequence is that absolute zero cannot be reached in a finite number of steps. The record for the lowest temperatures today is below a billionth of a kelvin in ultracold atomic gases – see Superconductivity and superfluidity.

Phase transitions

At a phase transition such as evaporation the two phases have the same Gibbs energy per molecule. This gives the Clausius–Clapeyron equation for how the boiling point depends on pressure:

$$\frac{dp}{dT} = \frac{L}{T\,\Delta v} \quad \Rightarrow \quad p(T) \approx p_0 \exp\left[-\frac{L_m}{R}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right]$$

That is why water boils at 70 °C on top of Mount Everest, and why a pressure cooker is faster. For ice and water Δv is negative – ice takes up more room than water – so the melting point falls slightly with pressure.



Boiling point

Denmark	100.0 °C
Ejer Bavnehøj	99.4 °C
Zugspitze	90.4 °C
Mont Blanc	84.7 °C
Kilimanjaro	81.4 °C
Mount Everest	72.9 °C

In 1869 Thomas Andrews discovered the critical point of carbon dioxide, and Johannes van der Waals explained it in his doctoral thesis of 1873 with a simple equation for real gases – see Properties of matter. In 1933 Paul Ehrenfest classified phase transitions according to whether the first derivatives (such as entropy and volume) or only higher derivatives of the Gibbs energy jump. Second-order transitions, as in a magnet at the Curie temperature or in superfluid helium at the λ point, are treated in Statistical mechanics.

Refrigerators and heat pumps

Run a heat engine backwards, and it moves heat from cold to hot – which requires work, as Clausius's statement says. A refrigerator moves heat out of the cold compartment; a heat pump moves heat from the outside air or the ground into a house. Its *coefficient of performance* (COP) is the heat delivered divided by the electricity used, and has the ideal limit

$$\mathrm{COP}_{\max} = \frac{T_h}{T_h - T_c}$$

With 35 °C in the radiators and 0 °C outside the limit is $308/35 \approx 8.8$; real air-to-water heat pumps typically achieve 3–4. Each kilowatt-hour of electricity thus gives 3–4 kilowatt-hours of heat, because the rest is taken from the surroundings. That is why heat pumps, and large heat pumps for district heating, are a central part of the green transition in Denmark.

Further reading

- Mark W. Zemansky and Richard H. Dittman: *Heat and Thermodynamics* (7th ed. 1997)
- Herbert B. Callen: *Thermodynamics and an Introduction to Thermostatistics* (2nd ed. 1985)
- Peter Atkins: *The Laws of Thermodynamics. A Very Short Introduction* (Oxford 2010)
- Sadi Carnot: *Reflections on the Motive Power of Fire* (Dover)

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